DATA LINEAGE DOCUMENT & POWER BI IMPLEMENTATION GUIDE

Sprint 3 — Data Governance, Lineage & Analytics

Prepared from the CSV exports provided for the project. This document distinguishes observed file contents from assumptions that must be confirmed against the live PostgreSQL database and Pentaho jobs.

1. Purpose and Scope

This document records how procurement data moves from source/staging extracts into warehouse dimensions and fact tables, and provides step-by-step instructions to build the Sprint 3 Power BI dashboard. It covers purchase orders, suppliers, contracts, warehouses, inventory, invoices and dates.

Evidence boundary: the provided CSV files confirm the column names and current row counts of those exports. They do not, by themselves, prove which Pentaho job produced each file, the exact transformation sequence, live database connectivity, or successful production execution. Verify those items with the team before final sign-off.

2. High-Level Data Flow

Layer	Lineage description
1. Source files	Excel/CSV/JSON/XML procurement source data
2. Ingestion	Pentaho transformations read source files and load staging
3. Staging extracts	staging.po, staging.suppliers, staging.contracts, staging.warehouses, staging.inventory, staging.invoices
4. Transform / validate	Standardize data, convert dates/types, validate keys and business rules; exact steps must match .ktr files
5. Warehouse	Dimensions: dim_purchase_order, dim_supplier, dim_contract, dim_date, dim_warehouse; Facts: fact_purchase_order, fact_inventory, fact_invoice
6. Analytics	Power BI reports for procurement, supplier performance, inventory and invoice status

3. File Inventory and Observed Row Counts

CSV export	Rows in provided file	Example columns
staging_po	25000	po_id, po_date, supplier_id, product_id, warehouse_id ...
staging_suppliers	1750	supplier_id, supplier_name, supplier_type, supplier_country, supplier_city ...
staging_warehouses	10	warehouse_id, warehouse_name, city, country, warehouse_type ...
staging_contracts	600	contract_id, supplier_id, contract_name, start_date, end_date ...
staging_inventory	5000	inventory_id, po_id, product_id, warehouse_id, snapshot_date ...
staging_invoices	5000	invoice_id, po_id, supplier_id, invoice_date, receipt_date ...
dim_purchase_order	5000	po_key, po_id, product_id, supply_type, currency ...
dim_supplier	250	supplier_key, supplier_id, supplier_name, supplier_type, supplier_country ...
dim_warehouse	10	warehouse_key, warehouse_id, warehouse_name, city, country ...

dim_contract	600	contract_key, contract_id, supplier_id, contract_name, start_date ...
dim_date	915	date_key, full_date, day, month, month_name ...
fact_purchase_order	15000	po_fact_key, po_key, supplier_key, contract_key, warehouse_key ...
fact_inventory	5000	inventory_fact_key, inventory_id, po_key, warehouse_key, snapshot_date_key ...
fact_invoice	5310	invoice_fact_key, invoice_id, po_key, supplier_key, invoice_date_key ...

Row counts above are counts in the supplied CSV exports, not proof of the current live PostgreSQL row counts. The full column lists are documented in Section 4.

4. Dataset and Table Schemas

Staging: staging.po

`po_id`, `po_date`, `supplier_id`, `product_id`, `warehouse_id`, `contract_id`, `supply_type`, `quantity`, `subtotal_inr`, `freight_inr`, `import_duty_percent`, `import_duty_inr`, `tax_inr`, `total_po_value_inr`, `currency`, `exchange_rate_to_inr`, `shipping_mode`, `expected_delivery_date`, `actual_receipt_date`, `delay_days`, `approval_status`, `po_status`, `unit_cost_inr`

Staging: staging.suppliers

`supplier_id`, `supplier_name`, `supplier_type`, `supplier_country`, `supplier_city`, `primary_category`, `supplier_rating`, `typical_lead_time_days`, `supplier_status`, `currency`

Staging: staging.warehouses

`warehouse_id`, `warehouse_name`, `city`, `country`, `warehouse_type`, `capacity_units`, `storage_condition`, `warehouse_status`

Staging: staging.contracts

`contract_id`, `supplier_id`, `contract_name`, `start_date`, `end_date`, `contract_value`, `currency`, `contract_type`, `contract_status`, `payment_terms_days`

Staging: staging.inventory

`inventory_id`, `po_id`, `product_id`, `warehouse_id`, `snapshot_date`, `opening_stock`, `stock_in`, `stock_out`, `closing_stock`, `reorder_level`, `safety_stock`, `inventory_value_inr`, `stock_status`, `capacity_utilization_percent`

Staging: staging.invoices

`invoice_id`, `po_id`, `supplier_id`, `invoice_date`, `receipt_date`, `due_date`, `invoice_amount_inr`, `currency`, `exchange_rate_to_inr`, `payment_status`, `paid_date`, `matching_status`, `invoice_approval_status`

Observed CSV row count: 5000

Dimension: dw.dim_purchase_order

`po_key`, `po_id`, `product_id`, `supply_type`, `currency`, `shipping_mode`, `approval_status`, `po_status`

Observed CSV row count: 5000

Dimension: dw.dim_supplier

`supplier_key`, `supplier_id`, `supplier_name`, `supplier_type`, `supplier_country`, `supplier_city`, `primary_category`, `supplier_rating`, `typical_lead_time_days`, `supplier_status`, `currency`

Observed CSV row count: 250

Dimension: dw.dim_warehouse

`warehouse_key`, `warehouse_id`, `warehouse_name`, `city`, `country`, `warehouse_type`, `capacity_units`, `storage_condition`, `warehouse_status`

Observed CSV row count: 10

Dimension: dw.dim_contract

`contract_key`, `contract_id`, `supplier_id`, `contract_name`, `start_date`, `end_date`, `contract_value`, `currency`, `contract_type`, `contract_status`, `payment_terms_days`

Observed CSV row count: 600

Dimension: dw.dim_date

`date_key`, `full_date`, `day`, `month`, `month_name`, `quarter`, `year`

Observed CSV row count: 915

Fact: dw.fact_purchase_order

`po_fact_key`, `po_key`, `supplier_key`, `contract_key`, `warehouse_key`, `po_date_key`, `expected_delivery_key`, `actual_receipt_key`, `quantity`, `unit_cost_inr`, `subtotal_inr`, `freight_inr`, `import_duty_percent`, `import_duty_inr`, `tax_inr`, `total_po_value_inr`, `exchange_rate_to_inr`, `delay_days`

Observed CSV row count: 15000

Fact: dw.fact_inventory

`inventory_fact_key`, `inventory_id`, `po_key`, `warehouse_key`, `snapshot_date_key`, `product_id`, `opening_stock`, `stock_in`, `stock_out`, `closing_stock`, `reorder_level`, `safety_stock`, `inventory_value_inr`, `capacity_utilization_percent`

Observed CSV row count: 5000

Fact: dw.fact_invoice

`invoice_fact_key`, `invoice_id`, `po_key`, `supplier_key`, `invoice_date_key`, `receipt_date_key`, `due_date_key`, `paid_date_key`, `invoice_amount_inr`, `exchange_rate_to_inr`, `currency`, `payment_status`, `matching_status`, `invoice_approval_status`

Observed CSV row count: 5310

5. Source-to-Target Lineage Register

Source	Staging evidence	Dimension target	Fact target	Lineage meaning
purchase_orders.xlsx / purchase-order source	staging.po (CSV export)	dw.dim_purchase_order	dw.fact_purchase_order	PO identifiers and descriptive attributes feed the PO dimension; order measures and keys feed the PO fact.
suppliers.csv	staging.suppliers (CSV export)	dw.dim_supplier	dw.fact_purchase_order and dw.fact_invoice via supplier_key	Supplier attributes are represented in the supplier dimension; facts reference supplier key.

warehouses.json	staging.warehouses (CSV export)	dw.dim_warehouse	dw.fact_purchase_order and dw.fact_inventory via warehouse_key where present	Warehouse descriptive attributes feed the warehouse dimension.
contracts.csv	staging.contracts (CSV export)	dw.dim_contract	dw.fact_purchase_order via contract_key	Contract attributes feed the contract dimension; PO facts use contract key when a contract exists.
inventory.json	staging.inventory (CSV export)	dimtinventory	dw.fact_inventory	Inventory movement and stock snapshot measures load with PO, warehouse and snapshot date keys.
invoices.xml	staging.invoices (CSV export)		dw.fact_invoice	Invoice records load with PO, supplier and invoice/receipt/due/paid date keys.
Date generation/source process	Not represented by a source staging export in supplied files	dw.dim_date	Date keys in all applicable fact tables	Date dimension maps date values to date_key for reporting and time analysis.

6. Key Relationships for the Power BI Model

Dimension key	Fact key	Purpose / caution
dw.dim_purchase_order[po_key]	dw.fact_purchase_order[po_key]	Purchase-order descriptive dimension to PO fact
dw.dim_purchase_order[po_key]	dw.fact_inventory[po_key]	Inventory records to purchase order
dw.dim_purchase_order[po_key]	dw.fact_invoice[po_key]	Invoices to purchase order
dw.dim_supplier[supplier_key]	dw.fact_purchase_order[supplier_key]	Supplier analysis of POs
dw.dim_supplier[supplier_key]	dw.fact_invoice[supplier_key]	Supplier invoice/payment analysis
dw.dim_warehouse[warehouse_key]	dw.fact_inventory[warehouse_key]	Inventory by warehouse
dw.dim_warehouse[warehouse_key]	dw.fact_purchase_order[warehouse_key]	PO warehouse analysis; verify actual fact column and key
dw.dim_contract[contract_key]	dw.fact_purchase_order[contract_key]	Contract compliance analysis
dw.dim_date[date_key]	fact date-key columns	Date analysis; role-playing date relationships need deliberate setup

Modeling note: Power BI can create ambiguity when one date dimension is connected to multiple date columns in a fact table. Start with one active date relationship per fact and add inactive relationships or duplicate role-playing date dimensions when needed. Do not connect fact tables directly unless there is a specific, justified modeling requirement.

7. Field-Level Lineage Examples

Source field	Target field	Rule / note
staging.po.po_id	dw.dim_purchase_order.po_id	Carry source PO identifier into PO dimension; validate uniqueness.
staging.po.product_id	dw.dim_purchase_order.product_id	Carry product identifier as a descriptive attribute; not a confirmed standalone product dimension.
staging.po.supply_type	dw.dim_purchase_order.supply_type	Carry supply type as a PO attribute.
staging.po.po_status	dw.dim_purchase_order.po_status	Carry status attribute; preserve approved status vocabulary.
staging.po.quantity	dw.fact_purchase_order.quantity	Numeric measure; validate null/negative values per agreed business rule.
staging.po.delay_days	dw.fact_purchase_order.delay_days	Delay measure; verify whether source or ETL derives it and confirm sign convention.
staging.inventory.opening_stock / stock_in / stock_out	dw.fact_inventory corresponding fields	Validate closing_stock = max(opening_stock + stock_in - stock_out, 0), if this is the agreed rule.
staging.invoices.payment_status	dw.fact_invoice.payment_status	Carry payment status; define allowed values with the team.
staging.invoices.paid_date	dw.fact_invoice.paid_date_key	Convert paid date to date_key; handle missing paid dates as null/unknown according to the agreed model.

8. Data Quality and Business Validation

Validation area	Suggested check	Evidence to save
Procurement transactions	PO identifiers are unique in the dimension; fact PO keys resolve; required date/supplier/warehouse keys resolve; quantities and date logic are sensible.	SQL or Power BI validation results; exceptions documented.
Supplier payments	Invoice identifiers and PO/supplier keys resolve; payment status is in the agreed list; paid date is not earlier than invoice date unless explained; overdue logic excludes paid invoices.	Invoice validation query and exception count.
Inventory reconciliation	Compare closing_stock with the agreed stock movement equation; stock quantities non-negative where required; capacity utilization within 0–100% if this is the agreed rule.	Rows checked, exceptions, and result evidence.
Contract compliance	Identify POs with missing contract references; compare PO date with contract start/end dates where business rules require it.	Exception list and agreed handling for no-contract POs.
Completeness / referential integrity	Check required fields, nulls, duplicates and unmatched foreign keys.	Record counts and failed key counts.

9. Power BI — Step-by-Step Instructions

Phase A: Prepare the data

1. Install Power BI Desktop from the official Microsoft page: https://powerbi.microsoft.com/desktop/ .
2. Use the data files supplied by the project team to build the report locally.
3. The report should be built from the supplied project data files; no live connection is required for this workflow.
4. Save a new report as EPDP_Sprint3_Executive_Dashboard.pbix.

Phase B: Load the supplied data files

5. In Power BI Desktop, choose Home → Get data, then select the appropriate file import option.
6. Load these dimension files: dim purchase order, dim supplier, dimcontract, dimdate, and dimwarehouse.
7. Load these fact files: factpurchaseorder, factinventory, and factinvoice.
8. Use the preview window to confirm headers, then choose Transform Data if you need to set data types.
9. In Power Query, set date columns to Date, key columns to Whole Number or Text according to their values, quantities/counts to Whole Number, percentages to Decimal Number, and status/name columns to Text. Keep IDs as text if they contain leading zeros.
10. Select Close & Apply.

Phase C: Build the data model

11. Open Model view. Create the relationships in Section 6 using the exact column names and data types.
12. Confirm that dimension-side keys are unique and that fact-side keys can repeat.
13. Use one-to-many relationships from dimensions to facts, with single-direction filtering initially.
14. Connect dim_date to one selected date key per fact table as the active relationship. If multiple date roles are required, create inactive relationships or separate role-playing date tables deliberately.
15. Check for blank/unmatched keys and avoid creating many-to-many relationships just to remove an error.

Phase D: Create starter measures

Total Purchase Orders

```
Total Purchase Orders = DISTINCTCOUNT(factpurchaseorder[po_key])
```

Counts distinct PO surrogate keys in the fact export; verify the key is populated.

Total Quantity

```
Total Quantity = SUM(factpurchaseorder[quantity])
```

Summed order quantity; not a currency metric.

Average Delivery Delay (Days)

```
Average Delivery Delay (Days) = AVERAGE(factpurchaseorder[delay_days])
```

Only use after confirming the sign convention and meaning of delay_days.

Invoice Count

```
Invoice Count = DISTINCTCOUNT(factinvoice[invoice_id])
```

Requires invoice export to represent current, complete data.

Invoices Pending Payment

```
Invoices Pending Payment = CALCULATE([Invoice Count], factinvoice[payment_status] <> "Paid")
```

Adjust the condition to the actual payment status vocabulary; handle blanks explicitly.

Phase E: Build three report pages

Report page	Suggested visuals
1. Executive Overview	Card: Total Purchase Orders; Card: Total Quantity; bar chart: PO count by PO status; column chart: PO count by supplier; line chart: PO count over PO date.
2. Supplier Performance	Bar chart: PO count by supplier; table: supplier name, rating, typical lead time; chart: average delay by supplier. Define on-time delivery as actual receipt date <= expected delivery date and exclude missing dates/cancelled orders according to agreed rules.
3. Inventory & Invoice Monitoring	Bar chart: closing stock by warehouse; table or card: inventory rows below reorder level; line chart: stock movement by snapshot date; stacked chart: invoice count by payment status; table: POs with missing contract key.

Design guidance: use a white background, a restrained corporate color palette, clear titles, consistent units and slicers for date, supplier, warehouse and status where appropriate. Do not display financial totals until the team has verified the source fields, currency handling and aggregation rules.

Phase F: Validate the report

16. Compare Power BI row counts with the supplied data files and investigate any differences against the team-provided reference values.
17. Test a few records against source/staging values.
18. Check that slicers filter all intended visuals and that date relationships behave correctly.
19. Document missing dates, unmatched keys, duplicate IDs and any incomplete fact-table export.
20. Save the PBIX, export a PDF if required, and capture screenshots for the Sprint 3 presentation.

10. Evidence and Handover Checklist

Deliverable	Acceptance evidence	Status
Source-to-target mapping	Mapping workbook reviewed and source/target names verified	Pending team sign-off
Data lineage	This document updated with actual .ktr/.kjb filenames and execution evidence	Working draft
Business glossary	Terms and KPI definitions agreed by business/team	Review required
ETL orchestration	Master .kjb runs in order and reports failures	Verify in Spoon
Validation	Procurement, invoice/payment and inventory checks saved with results	Run and attach evidence
Power BI	PBIX saved; relationships and measures checked; screenshots captured	Build the report using the supplied project data files; save the report and capture screenshots.
Git release	Final commit and version tag created after team review	Pending
Report/presentation	Include architecture, lineage, validation, dashboards, limitations and next steps	Pending

11. Assumptions and Open Questions

- The supplied files are used as the project reference for this document; confirm any unresolved lineage details with the team.
- Exact Pentaho transformation/job filenames and actual execution order were not inferable from the supplied CSV data alone.
- The row counts recorded in this document describe the supplied project files.
- Confirm the date dimension generation process and how date roles are modeled in Power BI.
- Confirm business definitions for on-time delivery, late invoice, pending payment, contract compliance and inventory reconciliation.
- Confirm whether monetary fields should be included in the executive dashboard and which currency/FX rules apply.

Appendix — Actual Column Lists

Staging: staging.po
po_id, po_date, supplier_id, product_id, warehouse_id, contract_id, supply_type, quantity, subtotal_inr, freight_inr, import_duty_percent, import_duty_inr, tax_inr, total_po_value_inr, currency, exchange_rate_to_inr, shipping_mode, expected_delivery_date, actual_receipt_date, delay_days, approval_status, po_status, unit_cost_inr

Staging: staging.suppliers
supplier_id, supplier_name, supplier_type, supplier_country, supplier_city, primary_category, supplier_rating, typical_lead_time_days, supplier_status, currency

Staging: staging.warehouses
warehouse_id, warehouse_name, city, country, warehouse_type, capacity_units, storage_condition, warehouse_status

Staging: staging.contracts
contract_id, supplier_id, contract_name, start_date, end_date, contract_value, currency, contract_type, contract_status, payment_terms_days

Staging: staging.inventory
inventory_id, po_id, product_id, warehouse_id, snapshot_date, opening_stock, stock_in, stock_out, closing_stock, reorder_level, safety_stock, inventory_value_inr, stock_status, capacity_utilization_percent

Staging: staging.invoices
invoice_id, po_id, supplier_id, invoice_date, receipt_date, due_date, invoice_amount_inr, currency, exchange_rate_to_inr, payment_status, paid_date, matching_status, invoice_approval_status

Dimension: dw.dim_purchase_order
po_key, po_id, product_id, supply_type, currency, shipping_mode, approval_status, po_status

Dimension: dw.dim_supplier
supplier_key, supplier_id, supplier_name, supplier_type, supplier_country, supplier_city, primary_category, supplier_rating, typical_lead_time_days, supplier_status, currency

Dimension: dw.dim_warehouse
warehouse_key, warehouse_id, warehouse_name, city, country, warehouse_type, capacity_units, storage_condition, warehouse_status

Dimension: dw.dim_contract
contract_key, contract_id, supplier_id, contract_name, start_date, end_date, contract_value, currency, contract_type, contract_status, payment_terms_days

Dimension: dw.dim_date
date_key, full_date, day, month, month_name, quarter, year

Fact: dw.fact_purchase_order
po_fact_key, po_key, supplier_key, contract_key, warehouse_key, po_date_key, expected_delivery_key, actual_receipt_key, quantity, unit_cost_inr, subtotal_inr, freight_inr, import_duty_percent, import_duty_inr, tax_inr, total_po_value_inr, exchange_rate_to_inr, delay_days

Fact: dw.fact_inventory
inventory_fact_key, inventory_id, po_key, warehouse_key, snapshot_date_key, product_id, opening_stock, stock_in, stock_out, closing_stock, reorder_level, safety_stock, inventory_value_inr, capacity_utilization_percent

Fact: dw.fact_invoice
invoice_fact_key, invoice_id, po_key, supplier_key, invoice_date_key, receipt_date_key, due_date_key, paid_date_key, invoice_amount_inr, exchange_rate_to_inr, currency, payment_status, matching_status, invoice_approval_status